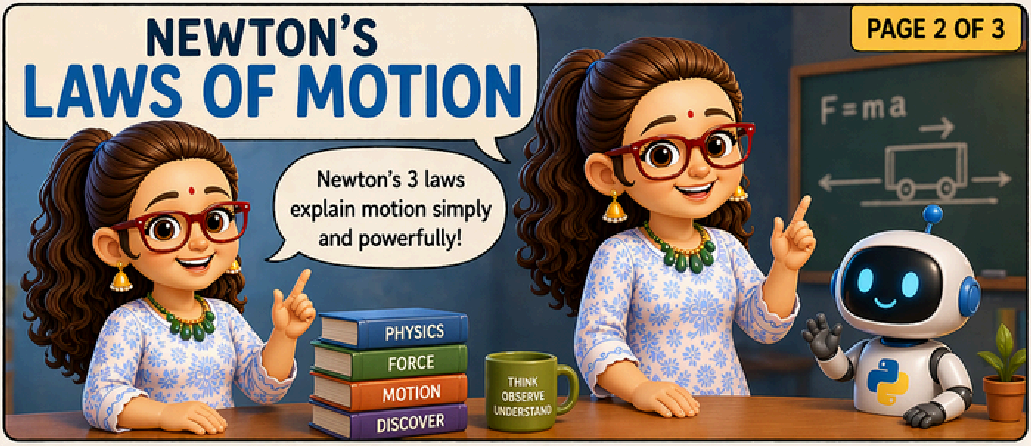

PHYSICS MADE EASY!
UNDERSTAND. EXPLORE. SUCCEED!
THE PHYSICS COMIC
VOL. 1
3 PAGES OF FUN & LEARNING!
WITH YOUR AI STUDY BUDDY!
LEARN • VISUALIZE • APPLY
INSIDE:
MOTION
FORCE
ENERGY
INTERACTIONS
...AND MORE!
COMPLEX TOPICS... SIMPLE EXPLANATIONS!
LET'S EXPLORE TOGETHER!
E=mc²
F=ma
PHYSICS
THE KEY TO UNDERSTANDING THE UNIVERSE
MECHANICS
THERMODYNAMICS
ELECTROMAGNETISM
OPTICS
MODERN PHYSICS
PERFECT FOR
✓ STUDENTS
✓ TEACHERS
✓ CURIOUS MINDS
✓ FUTURE SCIENTISTS!
PHYSICS ISN'T JUST A SUBJECT... IT'S HOW THE WORLD WORKS!
REAL EXAMPLES!
EASY DIAGRAMS!
QUICK RECAPS!
FUN TO LEARN!


PAGE 1 OF 3
Let's make PHYSICS EASY & FUN!
Let's start with 4 big ideas.
Physics is all about how the universe works!
E = mc²
MECHANICS
THERMODYNAMICS
ELECTROMAGNETISM
OPTICS
THINK OBSERVE UNDERSTAND
1. MOTION
Objects move and change position.
$v = \frac{d}{t}$
Speed = $\frac{\text{Distance}}{\text{Time}}$
2. FORCE
A push or pull can change motion.
F = m × a
Force = mass × acceleration
3. ENERGY
Energy can't be created or destroyed.
Energy changes form, not amount.
4. INTERACTION
Everything in the universe is connected.
Gravitation, friction, electricity... all link one thing to another.
We'll explore each idea step by step with examples, diagrams, and real-life fun!
Can't wait to learn!
PHYSICS FUNDAMENTALS
THE LAWS OF NATURE
QUESTIONS. EXPLORE. DISCOVER.
NEXT PAGES PREVIEW (PROMPTS)
PAGE 2: NEWTON'S LAWS OF MOTION
• Law 1: Inertia (Object at rest stays at rest)
• Law 2: F = m × a (Force causes acceleration)
• Law 3: Action = Reaction
• Real-life examples (car, football, rocket)
• Mini question box
PAGE 3: ENERGY, WORK & POWER
• Kinetic vs Potential Energy
• Work = Force × Distance
• Power = Work / Time
• Real-life examples (fan, lift, cyclist)
• Quick recap + Think like a physicist!
PHYSICS ISN'T JUST A SUBJECT... IT'S HOW THE WORLD WORKS!

NEWTON'S LAWS OF MOTION



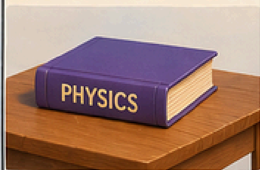
1. LAW OF INERTIA

An object at rest stays at rest, and an object in motion stays in motion with the same speed and direction, unless acted on by a net external force.

Objects resist changes in their state!

EXAMPLES

At rest, stays at rest.



The book won't move until a force acts on it.

In motion, stays in motion.

The car keeps moving at the same speed.

2. LAW OF ACCELERATION

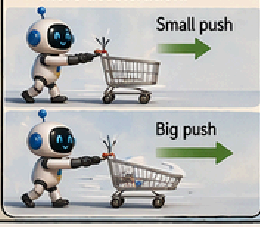
The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

$$F = m a$$

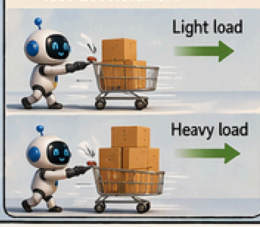
More force = more acceleration! More mass = less acceleration!

EXAMPLES

Same mass, more force = more acceleration.



More mass, same force = less acceleration.



3. LAW OF ACTION–REACTION

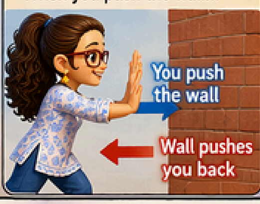
For every action, there is an equal and opposite reaction.

Action → Reaction

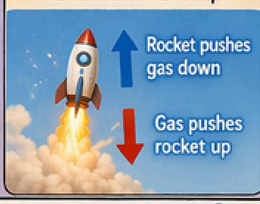
For every force, there's an equal and opposite force!

EXAMPLES

When you push the wall...



When a rocket moves up...



These 3 laws work together to explain everything around us!

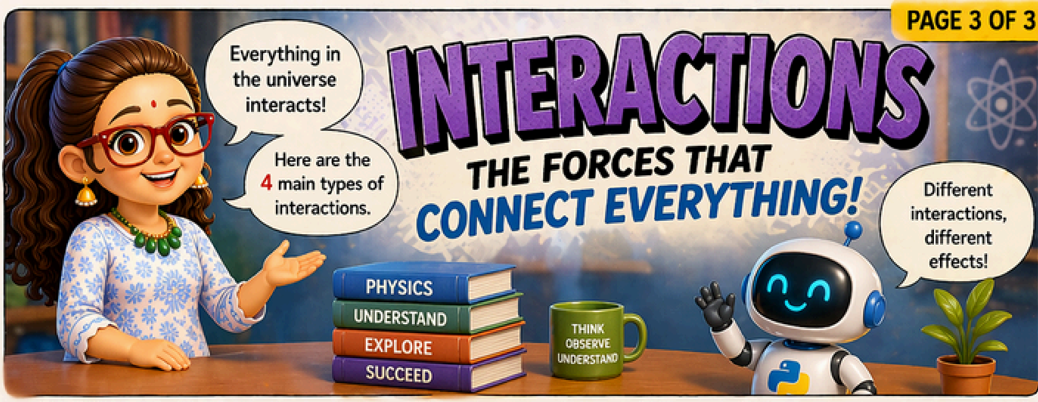
REMEMBER:

1. **Inertia** – Objects resist changes.
2. **Acceleration** – Force changes motion.
3. **Action–Reaction** – Every force has an equal and opposite force.

Physics makes the world make sense!

★ OBSERVE. THINK. UNDERSTAND. THAT'S THE SCIENTIFIC WAY! ★

PAGE 3 OF 3
ENERGY, WORK & POWER
Let's learn them one by one!
Three big ideas that make things happen!
ENERGY
WORK
POWER
PHYSICS
THINK OBSERVE UNDERSTAND
1. ENERGY
Energy is the ability to do work or cause change.
Types of Energy:
Kinetic (Movement)
Potential (Stored)
Thermal (Heat)
Electrical (Flow of charge)
Chemical (Stored in batteries, food, etc.)
EXAMPLES
A moving ball has kinetic energy.
A book on a shelf has potential energy.
PHYSICS
A hot cup of tea has thermal energy.
Energy can change from one form to another!
2. WORK
Work is done when a force causes displacement in the direction of the force.
No displacement means no work!
Formula:
W = F × d × cos θ
W = Work (Joules)
F = Force (Newtons)
d = Displacement (meters)
θ = Angle between force and displacement
EXAMPLES
You push a box 10 m with a force of 20 N.
W = F × d
= 20 × 10
= 200 J
You push a wall, but it doesn't move.
d = 0
W = 0 J
You push at an angle (θ = 60°) and move the box 5 m.
F
60°
d
W = F × d × cos 60°
= F × 5 × 0.5
= 2.5 F J
KEY IDEA
Work depends on force, distance, and direction!
Unit of Work: Joule (J)
3. POWER
Power is the rate at which work is done.
Do the same work faster, you have more power!
Formula:
P = W / t
P = Power (Watts)
W = Work (Joules)
t = Time (seconds)
EXAMPLES
Rhea lifts a bag (200 J) in 10 seconds.
P = 200 / 10
= 20 W
Aarav lifts the same bag (200 J) in 5 seconds.
P = 200 / 5
= 40 W
Both do the same work, but Aarav has more power!
More power = Faster work!
REAL-LIFE EXAMPLES
A car's engine does work to move the car (Work) and the rate at which it does it is power (Power).
A 60 W bulb uses 60 Joules of energy every second.
Cyclist doing more work uphill in the same time has more power!
Let's summarize!
ENERGY
✓ Ability to do work or cause change.
✓ Comes in many forms.
✓ Unit: Joule (J)
WORK
✓ Force causes displacement.
✓ Depends on force, distance & direction.
✓ Unit: Joule (J)
F
d
POWER
✓ Rate of doing work.
✓ More work in less time = more power.
✓ Unit: Watt (W)
Energy makes it possible, Work makes it happen, Power makes it impressive!
UNDERSTAND ENERGY. DO WORK. BE POWERFUL!



INTERACTIONS

THE FORCES THAT CONNECT EVERYTHING!

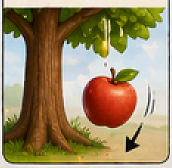
1. GRAVITATIONAL INTERACTION

Acts between objects that have mass. Always attractive.

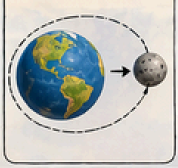
Keeps planets in orbit and objects falling to Earth.

EXAMPLES

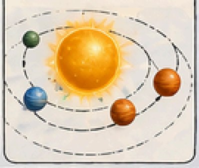
Apple falls to the ground.



The Moon orbits the Earth.

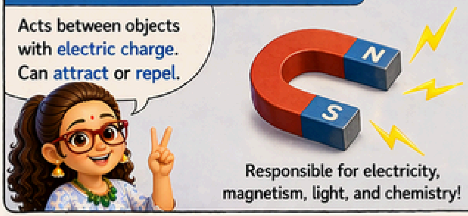


Planets orbit the Sun.



2. ELECTROMAGNETIC INTERACTION

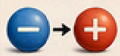
Acts between objects with electric charge. Can attract or repel.



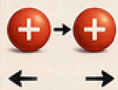
Responsible for electricity, magnetism, light, and chemistry!

EXAMPLES

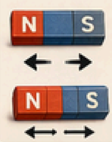
Opposite charges attract.



Like charges repel.



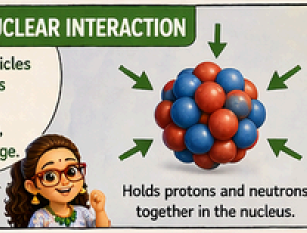
Magnets attract or repel.



Light is a form of electromagentic interaction.

3. STRONG NUCLEAR INTERACTION

Acts between particles inside the nucleus of an atom. Extremely strong, but very short range.

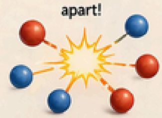


Holds protons and neutrons together in the nucleus.

EXAMPLES

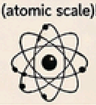
Keeps the nucleus stable.

Without it, the nucleus would fall apart!



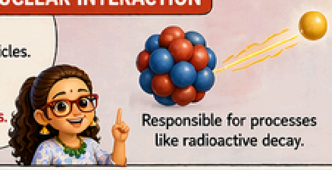
REMEMBER

It works only over tiny distances (atomic scale)!



4. WEAK NUCLEAR INTERACTION

Acts on certain subatomic particles. Very weak, but important for nuclear changes.



Responsible for processes like radioactive decay.

EXAMPLES

A neutron can change into a proton, emitting an electron.

Helps stars produce energy through nuclear reactions.

REMEMBER

It's rare, but crucial in many nuclear processes!

QUICK RECAP

INTERACTION	ACTS ON	NATURE	RANGE	EXAMPLE
Gravitational	Mass	Attractive	Infinite	Gravity, Orbits
Electromagnetic	Electric charge	Attracts or repels	Infinite	Magnets, Light
Strong Nuclear	Nucleons (in nucleus)	Very strong	Very short	Holds nucleus together
Weak Nuclear	Subatomic particles	Very weak	Very short	Radioactive decay

DIFFERENT INTERACTIONS, ONE CONNECTED UNIVERSE!


A Sriju Comic
Curiosity Lives Here!
Stories that smile back at you